Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: M.Tech(IIIrd)
Name of the Paper: Information Retrieval and Natural Language Processing
Paper Code: MCS 329
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. How has NLP transformed the field of artificial intelligence? Provide examples from real-world applications. (CO1)

OR

b. Explain how NLP is being applied in the healthcare industry. Provide examples of two specific applications. (CO1)

Q2. (10 Marks)

a. Discuss the limitations of using regular expressions for large-scale text classification in comparison to machine learning techniques. (CO1)

OR

b. Describe how inflectional and derivational morphemes change the meaning or function of a word. How do NLP systems handle these morphemes in tasks like part-of-speech tagging or word embedding generation? (CO2)

Q3. (10 Marks)

a. Explain the difference between rule-based and statistical approaches to Part-of-Speech (POS) tagging. (CO1)

OR

b. Natural languages have several unique properties. Explain how ambiguity and context-dependence affect the performance of NLP models. Provide examples to support your explanation. (CO2)

Q4. (10 Marks)

a. Describe how the Bag of Words model works with an example. What are the major limitations of the Bag of Words model, and how can these be mitigated in practice? (CO2)

OR

b. Write a regular expression that extracts all email addresses from a given text. Explain each part of the regular expression. (CO1)

Q5. (10 Marks)

a. What are some common tools used for syntactic and semantic analysis in NLP? Discuss their functionalities and provide examples of how these tools can be used for linguistic analysis in an NLP pipeline. (CO2)

OR

b. Explain the process of web scraping using Python. What libraries are commonly used for web scraping in Python, and how do they assist in extracting data from websites? Provide a code example that scrapes data from a webpage. (CO2)

Note For the question paper setters:

- Question paper should cover all the COs of the course.

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